

ACT-PKR-4500 'Patriot' Hydraulic Set Retrievable Packer

Technical Datasheet DS-PKR-4500 Rev C | Product family: ACT-PKR-4500

1. Product Overview

The ACT-PKR-4500 'Patriot' is a hydraulic set, single-string retrievable production packer for 4-1/2 in. casing. It sets with applied tubing pressure against a plugged tubing string, requires no tubing rotation, and releases with straight pull. A dual-slip system anchors the packer against pressure from above or below, and a three-piece element package provides a reliable long-term seal. The packer is qualified to API 11D1 Grade V3.

2. Applications

- Single-zone production or injection completions
- Wells requiring TCP or stimulation before release
- Deviated wells where rotation is impractical
- Water and CO2 injection service (Aflas option)

3. Features

- Sets with 2,950 psi applied tubing pressure - no rotation required
- Straight-pull release with adjustable shear release value
- Bi-directional wicker slips, 8620 carburized
- Three-piece HNBR element system, 300 degF standard (Aflas optional per ECO-2024-067)
- Full-bore ID compatible with standard 2-7/8 in. completion accessories
- API 11D1 Grade V3 qualified

4. Technical Specifications

Parameter	Specification
Casing Size	4-1/2 in., 9.5 - 13.5 lb/ft
Maximum Tool OD	3.780 in.
Minimum Bore ID	1.938 in.
Overall Length (assembled)	58.25 in.
Differential Pressure Rating	10,000 psi (from above or below)
Temperature Rating	300 degF standard (HNBR) / 350 degF (Aflas option)
Setting Pressure	2,950 +/- 150 psi applied tubing pressure
Release	Straight pull, 40,000 lbf over string weight (std shear ring)
End Connections	2-7/8 in. EUE 8RD box up x pin down (NUE and premium on request)
Tensile Rating (body)	185,000 lbf
Qualification	API 11D1 Grade V3; NACE MR0175 trim optional
Service	Standard (H2S trim optional, 13Cr/Aflas)

All ratings are at ambient temperature unless otherwise noted. Performance envelope per API 11D1 V3 qualification testing where indicated. Consult ACT Engineering for combined load conditions.

5. Materials of Construction

Component	Material
Mandrel / Subs	AISI 4140, 28-32 HRC
Slips	AISI 8620, carburized 58-62 HRC
Element (std)	HNBR 80/90/80 durometer, compound ACT-H90/H80
Element (option)	Aflas 90A, kit ACT-PKR-4500-03-AF
O-rings	Viton 75A

6. Ordering & Reference Documents

Order by part number and specify: casing size and weight, service trim (standard or NACE sour), elastomer option, end connections, and any customer inspection requirements. Related documents:

- BOM-PKR-4500 Rev C
- MAN-PKR-4500 Operations Manual
- TS-100 Torque Specification
- ECO-2024-067 (Aflas element option)
- HTR-2026-0044 (production hydrotest example)

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